हिन्दुस्तान पेट्रोलियम HP	Residue Upgradation Facility (RUF) EPCC-3 Package for Visakh Refinery Modernization Project (VRMP)	L&T Hydrocarbon Engineering
इंजीनियर्स इंडिया लिमिटेड (भारत सरकार का उपक्रम) ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)		**L&T- CHIYODA LIMITED**
Title: GAD For Dock Leveler (504-LZ-930 A/B) **Doc. No.: BO16-RUF-LT-504-MP-GN-NTPL(1)-13002**		**Rev. No.: 02** **Page No: 1 of 4**

OWNER : **HINDUSTAN PETROLEUM CORPORATION LIMITED (HPCL)**
PMC : **ENGINEERS INDIA LIMITED, NEW DELHI (EIL)**
UNIT : **504**
PMC JOB NO. : **B016**
EQUIPMENT DESCRIPTION : **DOCK LEVELER**
EQUIPMENT TAG : **504-LZ-930A/B**
P. R. NO. : **BO16-RUF-LT-504-MP-PR-0014**
VENDOR NAME : **NIHVA TECHNOLOGIES PVT LTD**
VENDOR DOCUMENT NO. : **NIHVA_20-21_0157**

DOCUMENT SUBMISSION STATUS – LTHE/VENDOR	**REVIEW STATUS**	
	LTHE	**EIL**
DOC. CATEGORY: (USE TICK MARK) ☑ R ☐ I	☐ Code 1 – No Comments. Proceed With Manufacture / Fabrication / Construction As Per The Document.	☐ Code 1 – No Comments. Proceed With Manufacture / Fabrication / Construction As Per The Document.
ISSUED FOR	☐ Code 2 – Proceed With Manufacture/Fabrication/Construction As Per Commented Document. Revised Document Required.	☐ Code 2 – Proceed With Manufacture/Fabrication/Construction As Per Commented Document. Revised Document Required.
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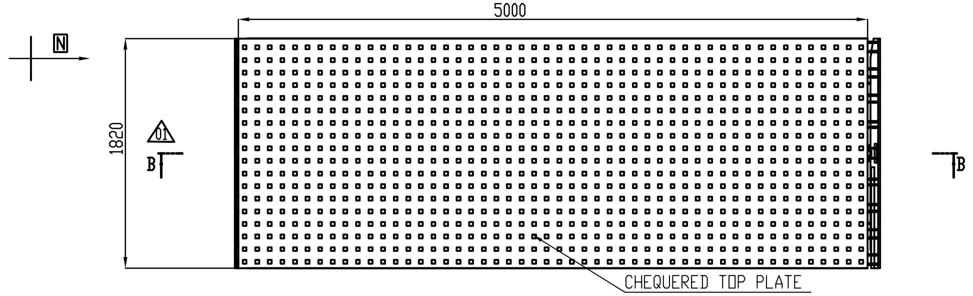
	DISCIPLINE | SIGN | DATE	DISCIPLINE | SIGN | DATE

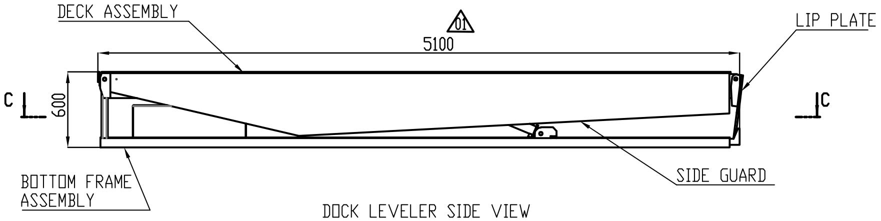
02	Issued for Approval	Shilpa	Amit K	Amit K			23/02/2022
01	Issued for Approval	Shilpa	Amit K	Amit K			24/09/2021
Revision No.	**Description**	**Prepared by**	**Reviewed by**	**Approved by**	**Reviewed by**	**Approved by**	**Approved Date**

REVISION HISTORY SHEET

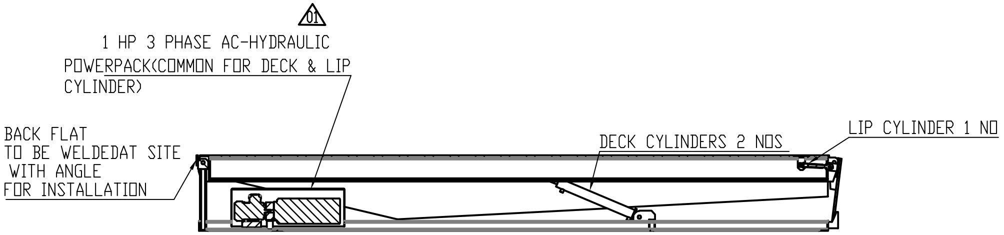
REV. NO.	REV. DATE	REVISED PAGE	REVISION DESCRIPTION
01	24/09/2021	NA	ISSUED FOR APPROVAL
02	23/02/2022	NA	ISSUED FOR APPROVAL



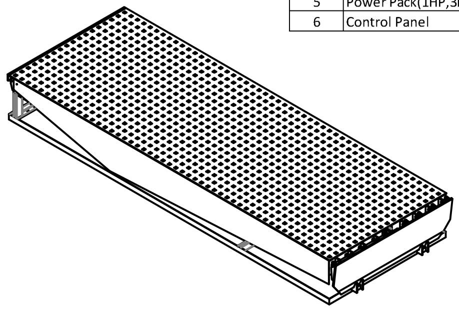
DOCK LEVELER TOP VIEW



DOCK LEVELER SIDE VIEW


△01
1 HP 3 PHASE AC-HYDRAULIC POWERPACK(COMMON FOR DECK & LIP CYLINDER)
BACK FLAT TO BE WELDEDAT SITE WITH ANGLE FOR INSTALLATION
DECK CYLINDERS 2 NOS
LIP CYLINDER 1 NO

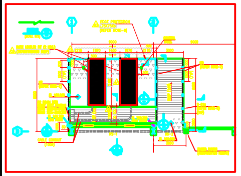
SECTION B-B
SCALE 1 : 15

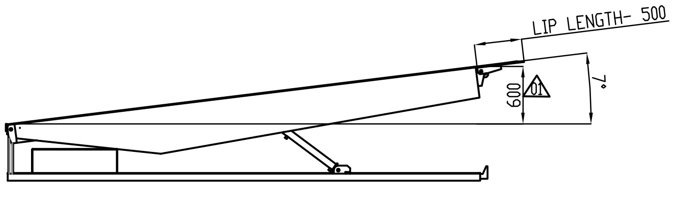


ISOMETRIC VIEW OF DOCK LEVELER

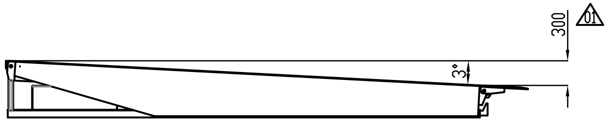
BOM (For Single Dock Leveler)		
SL NO.	DESCRIPTION	QTY
1	Dock Leveler(1820m x 5000mm)	1No
2	Lip (1820 x500mm)	1No
3	Deck Cylinder	2Nos
4	Lip Cylinder	1No
5	Power Pack(1HP,3Phase)	1No
6	Control Panel	1No

△02



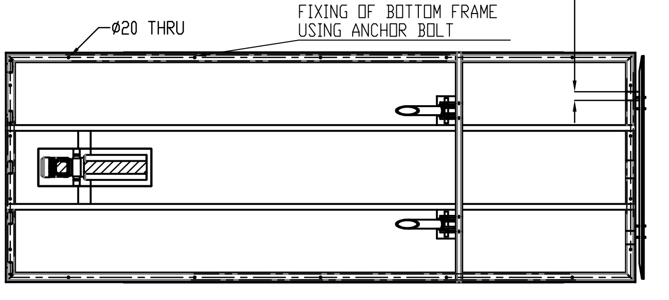


DECK UP HEIGHT - 600 MM



DECK DOWN HEIGHT - 300 MM △02

40MM CUTOUT AT THE CENTRE OF THE TUBE FOR WATER DRAINNG △02



SECTION C-C
SCALE 1 : 15

△01

TECHNICAL SPECIFICATIONS OF PIT MOUNTED ELECRO-HYDRAULIC DOCK LEVELER -

1. SIZE OF DOCK LEVELER = 1820mm(W) X 5000mm(L)
2. OVERALL SIZE OF PRODUCT = 1820mm(W)X5100mm(L)X600mm(D)
3. CAPACITY = 6 TON POINT LOAD
4. LIP PLATE SIZE = 500mm(L)X1820mm(W)
5. DECK PLATE = CHEQUERED SHEET 1820mm(W)x5000mm(L)
6. DECK & LIP UP=DOWN OPERATION BY HYDRAULIC CYLINDER
7. DECK UP HEIGHT= (+)600 MM
8. DECK DOWN HEIGHT= (-)300MM △02
9. HEIGHT OF DOCK LEVELER FROM ROAD LEVEL= 900 mm,
10. OPERATION CONTROL BY PUSH BUTTONS ON CONTROL PANEL MOUNTED ON STAND NEAR DOCK LEVELER
11. ELECTRICAL SUPPLY OF EQUIPMENT= 415 VOLT 3 PHASE AC , 50HZ
12. PAINT = FIROZI BLUE
13. MATERIAL OF CONSTRUCTION = MILD STEEL AS PER IS STANDARD
14. MANUFACTURING STANDARD OF DOCK LEVELER =EN1398
15. ELECRICAL MOTOR COMPLIANCE ZONE 1-IIC-T3 HAZARDOUS AREA
16. PIT SIZE = 1870 mm(W)X5100mm(L)x600 mm(D)
17. DOCK LEVELER FIXING IN PIT BY ANCHOR BOLT .
18. REQUIRED PIT & ALL CIVIL WORK FOR MOUNTING OD DOCK LEVELER IS IN LTHE SCOPE .
    REFER PIT DRAWING FOR CONSTRUCTION OF PIT B016-RUF-LT-504-MP-DW-NTPL(1)-13005
19. HYDRAULIC CYLINDERS & POWERPACK IS DESIGNED FOR LIFTING BARE DOCK ONLY.
20. MOTOR RATING=1HP 3 PHASE AC-HYDRAULIC
21. All ELECTRICAL EQUIPMENTS SHALL HAVE PAINT SHADE RAL 7031 △02
22. ALL FLAME PROOF CONTROL PANEL 360X360X210MM △02
23. WEIGHT= APPROX 2500KG EMPTTY & 8500KG OPERATING △02
24. WE WILL PROVIDE FRP CANOPY FOR LOCAL CONTROL PANEL. △02

504-LZ-930A/B	CIVIL DRAWING OF DOCK LEVELER △01	01
DRAWING NO.	DRAWING TITLE	REV

REFERENCE DOCUMENTS / DRAWINGS

DRG. CATEGORY: (USE TICK MARK) ☐ R ☐ I
SCH. DATE OF SUBMISSION
EIL REVIEW/APPROVAL CODE:
VPTL NO.:

DOCUMENT STATUS
STATEMENT OF SUB: 1st 2nd 3rd
☐ FINAL FOR CODE-1 ☑ AS BUILT

ISSUED FOR
☐ AS BUILT ☐ INFORMATION
☐ CONSTRUCTION ☐ REVIEW

SIGN: DATE:

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☐ REVIEW CODE 2 – PROCEED WITH MANUFACTURE / FABRICATION / CONSTRUCTION AS PER THE COMMENTED DOCUMENT. REVISED DOCUMENT REQUIRED
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☐ V-VOID
NAME DICIPLINE SIGN. DATE

REV.	DESCRIPTION	DRN	CHECKED	APPROVED	REVIEWED	APPROVED	DATE
01	ISSUED FOR APPROVAL	SMS	VJ	AK			

OWNER HINDUSTAN PETROLEUM CORPORATION LIMITED
VISAKH REFINERY MODERNIZATION PROJECT
EPCC-03 PACKAGE FOR RESIDUE UP-GRADATION FACILITY (RUF)

PMC : इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

LSTK CONTRACTOR: L&T Hydrocarbon Engineering

DETAILED ENGINEERING CENTER: L&T - CHIYODA LIMITED

L&T - CHIYODA LIMITED
☐ REVIEW CODE 1 – NO COMMENTS PROCEED WITH MANUFACTURE / FABRICATION/ CONSTRUCTION AS PER THE DOCUMENT.
☐ REVIEW CODE 2 – PROCEED WITH MANUFACTURE/ FABRICATION /CONSTRUCTION AS PER THE COMMENTED DOCUMENT. REVISED DOCUMENT REQUIRED.
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☐ V-VOID

COMMENTED BY	REVIEWED BY	APPROVED BY	DATE

THIS REVIEW / APPROVAL DOES NOT RELIEVE THE VENDOR / MANUFACTURER OF THE FULL RESPONSIBILITY AS TO DESIGN CORRECTNESS OF DIMENSIONS & COMPLIANCE WITHALL APPLICABLE REGULATIONS AND WITH THE REQUIREMENTS MADE IN PURCHASER SPECIFICATIONS.

VENDOR : NIHVA TECHNOLOGIES PVT LTD
VENDOR PO NO: REFINARY/75000-82256/VMM
VENDOR DWG. NO: NIHVA_20-21_0157
DATE 31.05.21 SCALE 1:1
TITLE: GAD FOR DOCK LEVELER (504-LZ-930A/B) △02
DEPT: DESIGN
DRN BY: SMS
CHK BY: VJ
APP BY: AK
PROJECT: L&T-RUF
PWO NO.:
DRG. NO.: [illegible]
SH.3 OF4
REV. 02
CAD REVISE THIS DRG. USING CAD SYSTEM ONLY FILE:

DATA SHEET FOR DOCK LEVELER (504-LZ-930A/B)

Sr No	Parameter	Remark
1	Required Capacity	06 Ton
2	Leveler operation mode	AC power pack hydraulically operated
3	Lip operation mode	hydraulic
4	Operation control	by push buttons and control panel
5	Height of dock leveler from road Level	900 mm
6	Travel above & below dock level	600 mm & 300 mm
7	Platform Size (W x L)	1820 mm x 5000 mm
8	Lip size (W x L)	1820 mm x 500 mm
9	Maximum slope can be achieved	7 Degree
10	Minimum slope can be achieved	3 Degree

* Please refer Attachment-1 for standard catalog of Dock Leveler.